tau+ (-15)
Particle 0
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (155.351, 78.012, 64.259, 185.343)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 280 simE=120.101166
subgraph recHits: 0 missing=280 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 16 dE=0.001446
raw GEN: GenParticle #17 key=2
raw SIM: n/a

Primary
Vertex 9
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.291, 0.145, -4.371, 0.179)
nIn: 1 nOut: 3
raw GEN: GenVertex #11 key=-11
raw SIM: SimVertex #77 key=9

tau- (15)
Particle 1
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-155.351, -78.012, -64.259, 185.343)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.354602
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 15 dE=0.001331
raw GEN: GenParticle #18 key=3
raw SIM: n/a

Primary
Vertex 6
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.923, -0.464, -4.873, 0.204)
nIn: 1 nOut: 3
raw GEN: GenVertex #8 key=-8
raw SIM: SimVertex #74 key=6

tau+ (-15)
Particle 2
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (51.188, 13.788, 35.785, 63.984)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 186 simE=15.756817
subgraph recHits: 0 missing=186 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 16 dE=0.002110
raw GEN: GenParticle #19 key=4
raw SIM: n/a

Primary
Vertex 0
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.532, 0.143, -4.121, 0.189)
nIn: 1 nOut: 3
raw GEN: GenVertex #2 key=-2
raw SIM: SimVertex #68 key=0

tau- (15)
Particle 3
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-51.188, -13.788, -35.785, 63.984)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 500 simE=24.355600
subgraph recHits: 0 missing=500 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 113 dE=0.021377
raw GEN: GenParticle #20 key=5
raw SIM: n/a

Primary
Vertex 1
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.316, -0.086, -4.713, 0.180)
nIn: 1 nOut: 5
raw GEN: GenVertex #3 key=-3
raw SIM: SimVertex #69 key=1

tau+ (-15)
Particle 4
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (252.217, -205.919, 246.510, 408.394)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 888 simE=118.227020
subgraph recHits: 0 missing=888 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 25 dE=0.002908
raw GEN: GenParticle #21 key=6
raw SIM: n/a

Primary
Vertex 2
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.245, -0.199, -4.253, 0.180)
nIn: 1 nOut: 3
raw GEN: GenVertex #4 key=-4
raw SIM: SimVertex #70 key=2

tau- (15)
Particle 5
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-252.217, 205.919, -246.510, 408.394)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 260 simE=60.812279
subgraph recHits: 0 missing=260 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 21 dE=0.002307
raw GEN: GenParticle #22 key=7
raw SIM: n/a

Primary
Vertex 3
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.063, 0.053, -4.554, 0.170)
nIn: 1 nOut: 3
raw GEN: GenVertex #5 key=-5
raw SIM: SimVertex #71 key=3

tau+ (-15)
Particle 6
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-63.885, 9.016, 119.820, 136.098)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 425 simE=41.867741
subgraph recHits: 22 missing=403 recoE=5.407163
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=0.006185
raw GEN: GenParticle #23 key=8
raw SIM: n/a

Primary
Vertex 4
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.061, 0.009, -4.373, 0.171)
nIn: 1 nOut: 5
raw GEN: GenVertex #6 key=-6
raw SIM: SimVertex #72 key=4

tau- (15)
Particle 7
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (63.885, -9.016, -119.820, 136.098)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 141 simE=4.106602
subgraph recHits: 0 missing=141 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 19 dE=0.001894
raw GEN: GenParticle #24 key=9
raw SIM: n/a

Primary
Vertex 5
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.669, -0.094, -5.743, 0.214)
nIn: 1 nOut: 2
raw GEN: GenVertex #7 key=-7
raw SIM: SimVertex #73 key=5

tau+ (-15)
Particle 8
visibleTau
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (35.712, -31.504, -206.570, 211.996)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 341 simE=0.054505
subgraph recHits: 257 missing=84 recoE=35.930195
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 19 dE=0.001694
raw GEN: GenParticle #25 key=10
raw SIM: n/a

Primary
Vertex 7
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.174, -0.153, -5.491, 0.201)
nIn: 1 nOut: 2
raw GEN: GenVertex #9 key=-9
raw SIM: SimVertex #75 key=7

tau- (15)
Particle 9
visibleTau
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-35.712, 31.504, 206.570, 211.996)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1115 simE=0.421371
subgraph recHits: 872 missing=243 recoE=91.599129
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 30 dE=0.003417
raw GEN: GenParticle #26 key=11
raw SIM: n/a

Primary
Vertex 8
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.160, 0.142, -3.559, 0.199)
nIn: 1 nOut: 2
raw GEN: GenVertex #10 key=-10
raw SIM: SimVertex #76 key=8

anti-nu_tau (-16)
Particle 10
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (34.286, 9.126, 24.387, 43.052)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #27 key=12
raw SIM: n/a

pi0 (111)
Particle 11
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (11.397, 3.011, 8.112, 14.311)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 118 simE=14.126666
subgraph recHits: 0 missing=118 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #28 key=13
raw SIM: n/a

Primary
Vertex 10
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.532, 0.143, -4.120, 0.189)
nIn: 1 nOut: 2
raw GEN: GenVertex #12 key=-12
raw SIM: SimVertex #78 key=10

pi+ (211)
Particle 12
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (5.505, 1.651, 3.286, 6.621)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 87 simE=1.630152
direct recHits: 0 missing=87 recoE=0.000000
subgraph simHits: 87 simE=1.630152
subgraph recHits: 0 missing=87 recoE=0.000000
direct tracker simHits: 16 dE=0.002110
subgraph tracker simHits: 16 dE=0.002110
checkpointId: 0
x4@checkpoint: (121.531, 20.810, 66.171, 0.000)
p4@checkpoint: (5.710, 0.266, 3.269, 6.587)
raw GEN: GenParticle #29 key=14
raw SIM: SimTrack #269 key=2

nu_tau (16)
Particle 13
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-9.930, -2.906, -6.626, 12.287)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #30 key=15
raw SIM: n/a

pi0 (111)
Particle 14
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-9.689, -2.915, -6.693, 12.132)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 256 simE=11.431248
subgraph recHits: 0 missing=256 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 71 dE=0.013791
raw GEN: GenParticle #31 key=16
raw SIM: n/a

Primary
Vertex 11
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.316, -0.086, -4.713, 0.180)
nIn: 1 nOut: 2
raw GEN: GenVertex #13 key=-13
raw SIM: SimVertex #79 key=11

pi- (-211)
Particle 15
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-19.619, -5.148, -13.876, 24.576)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 11
nCheckpoints: 1
direct simHits: 257 simE=12.171839
direct recHits: 0 missing=257 recoE=0.000000
subgraph simHits: 298 simE=12.208804
subgraph recHits: 0 missing=298 recoE=0.000000
direct tracker simHits: 39 dE=0.004540
subgraph tracker simHits: 39 dE=0.004540
checkpointId: 0
x4@checkpoint: (-118.118, -35.369, -88.831, 0.000)
p4@checkpoint: (-19.177, -6.465, -13.820, 24.506)
raw GEN: GenParticle #32 key=17
raw SIM: SimTrack #270 key=4

HadronInelastic

Vertex 178
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-110.108, -35.170, -142.913, 12.302)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #261 key=193

HadronInelastic

Vertex 179
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-118.593, 12.436, -101.444, 11.285)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #262 key=194

HadronInelastic

Vertex 180
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.665, -106.699, -54.354, 35.876)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #263 key=195

HadronInelastic

Vertex 181
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.688, 26.638, -120.437, 20.156)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #264 key=196

HadronInelastic

Vertex 182
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.883, 32.305, -83.216, 30.352)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #266 key=198

Bremsstrahlung
Vertex 183
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-38.367, -10.540, -31.546, 1.772)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #267 key=199

Bremsstrahlung
Vertex 184
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-37.618, -10.507, -71.873, 4.241)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #268 key=200

pi- (-211)
Particle 16
stableLegsFromUpstream
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-3.723, -0.892, -2.516, 4.583)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 9 simE=0.301080
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 8 dE=0.000588
subgraph tracker simHits: 15 dE=0.001371
raw GEN: GenParticle #33 key=18
raw SIM: SimTrack #271 key=5

Decay
Vertex 177
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-41.574, -12.870, -33.194, 1.908)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #260 key=192

pi+ (211)
Particle 17
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-8.227, -1.926, -6.074, 10.407)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 19 simE=0.414457
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 19 simE=0.414457
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 17 dE=0.001674
subgraph tracker simHits: 17 dE=0.001674
checkpointId: 0
x4@checkpoint: (-121.971, -18.055, -93.106, 0.000)
p4@checkpoint: (-8.411, -0.525, -6.047, 10.374)
raw GEN: GenParticle #34 key=19
raw SIM: SimTrack #272 key=6

anti-nu_tau (-16)
Particle 18
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (150.581, -123.568, 147.234, 244.175)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #35 key=20
raw SIM: n/a

pi0 (111)
Particle 19
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (46.512, -37.244, 44.902, 74.610)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 243 simE=73.457191
subgraph recHits: 0 missing=243 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #36 key=21
raw SIM: n/a

Primary
Vertex 12
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.246, -0.199, -4.252, 0.180)
nIn: 1 nOut: 2
raw GEN: GenVertex #14 key=-14
raw SIM: SimVertex #80 key=12

pi+ (211)
Particle 20
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (55.124, -45.107, 54.374, 89.610)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 20
nCheckpoints: 1
direct simHits: 780 simE=44.731129
direct recHits: 0 missing=780 recoE=0.000000
subgraph simHits: 809 simE=44.769871
subgraph recHits: 0 missing=809 recoE=0.000000
direct tracker simHits: 24 dE=0.002518
subgraph tracker simHits: 25 dE=0.002908
checkpointId: 0
x4@checkpoint: (94.637, -79.036, 89.626, 0.000)
p4@checkpoint: (54.192, -46.173, 54.347, 89.567)
raw GEN: GenParticle #37 key=22
raw SIM: SimTrack #273 key=8

HadronInelastic

Vertex 166
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-41.192, -111.942, 39.334, 29.288)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #248 key=180

HadronInelastic

Vertex 167
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.643, -65.590, 136.451, 14.490)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #249 key=181

HadronInelastic

Vertex 168
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.201, -92.617, 108.162, 11.822)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #250 key=182

HadronInelastic

Vertex 170
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.442, 117.660, 108.133, 39.310)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #252 key=184

HadronInelastic

Vertex 171
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (88.008, -79.530, 84.997, 8.442)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #253 key=185

HadronInelastic

Vertex 172
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.364, -76.475, 82.637, 11.528)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #254 key=186

Annihilation

Vertex 173
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (111.629, -40.607, 94.412, 12.296)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #255 key=187

HadronInelastic

Vertex 174
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.197, -83.774, 120.707, 21.143)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #257 key=189

Annihilation

Vertex 175
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (102.520, -53.674, 124.531, 7.611)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #258 key=190

HadronInelastic

Vertex 176
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.936, 26.834, 164.657, 34.652)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #259 key=191

nu_tau (16)
Particle 21
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-202.590, 164.833, -197.294, 327.319)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #38 key=23
raw SIM: n/a

pi0 (111)
Particle 22
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-37.226, 31.010, -37.072, 61.005)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 234 simE=60.377750
subgraph recHits: 0 missing=234 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #39 key=24
raw SIM: n/a

Primary
Vertex 13
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.065, 0.054, -4.556, 0.171)
nIn: 1 nOut: 2
raw GEN: GenVertex #15 key=-15
raw SIM: SimVertex #81 key=13

pi- (-211)
Particle 23
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-12.401, 10.076, -12.144, 20.070)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=0.434519
direct recHits: 0 missing=28 recoE=0.000000
subgraph simHits: 28 simE=0.434519
subgraph recHits: 0 missing=28 recoE=0.000000
direct tracker simHits: 21 dE=0.002307
subgraph tracker simHits: 21 dE=0.002307
checkpointId: 0
x4@checkpoint: (-99.048, 73.433, -98.255, 0.000)
p4@checkpoint: (-13.216, 8.928, -12.121, 20.033)
raw GEN: GenParticle #40 key=25
raw SIM: SimTrack #274 key=10

anti-nu_tau (-16)
Particle 24
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-10.649, 1.516, 19.150, 21.964)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #41 key=26
raw SIM: n/a

pi0 (111)
Particle 25
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-18.841, 2.558, 35.649, 40.403)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 240 simE=38.890572
subgraph recHits: 0 missing=240 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #42 key=27
raw SIM: n/a

Primary
Vertex 14
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.062, 0.009, -4.372, 0.172)
nIn: 1 nOut: 2
raw GEN: GenVertex #16 key=-16
raw SIM: SimVertex #82 key=14

pi+ (211)
Particle 26
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-28.611, 4.307, 54.374, 61.593)
nProdVtx: 1 nDecayVtx: 18
nParents: 1 nChildren: 23
nCheckpoints: 1
direct simHits: 29 simE=0.251908
direct recHits: 5 missing=24 recoE=0.513259
subgraph simHits: 29 simE=0.251908
subgraph recHits: 5 missing=24 recoE=0.513259
direct tracker simHits: 17 dE=0.001847
subgraph tracker simHits: 17 dE=0.001847
checkpointId: 0
x4@checkpoint: (-121.447, 21.298, 227.173, 0.000)
p4@checkpoint: (-28.333, 5.663, 54.333, 61.538)
raw GEN: GenParticle #43 key=28
raw SIM: SimTrack #275 key=12

HadronInelastic

Vertex 121
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-293.884, 61.822, 559.275, 21.476)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #203 key=135

HadronInelastic

Vertex 122
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-332.600, 78.315, 641.671, 24.563)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #204 key=136

PairConversion

Vertex 123
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-364.287, 51.804, 694.415, 26.801)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #205 key=137

Compton
Vertex 124
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-344.029, 60.208, 677.096, 25.951)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #206 key=138

PairConversion

Vertex 127
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-358.208, 102.940, 708.571, 27.096)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #209 key=141

PairConversion

Vertex 128
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-334.788, 79.093, 645.910, 24.724)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #210 key=142

PairConversion

Vertex 145
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-295.277, 61.611, 561.288, 21.558)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #227 key=159

HadronInelastic

Vertex 146
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-349.745, 74.281, 646.298, 24.951)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #228 key=160

Compton
Vertex 147
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-323.497, 62.913, 684.881, 26.553)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #229 key=161

Compton
Vertex 148
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-348.211, 265.670, 705.982, 31.654)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #230 key=162

PairConversion

Vertex 149
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-365.346, 121.560, 695.026, 27.275)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #231 key=163

HadronInelastic

Vertex 150
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-353.464, 74.992, 649.689, 25.123)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #232 key=164

PairConversion

Vertex 151
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-350.289, 74.299, 647.201, 24.986)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #233 key=165

PairConversion

Vertex 157
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-350.656, 74.260, 647.581, 25.004)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #239 key=171

Compton
Vertex 160
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-407.295, 102.623, 684.886, 27.448)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #242 key=174

PairConversion

Vertex 161
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-351.058, 74.135, 648.984, 25.051)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #243 key=175

HadronInelastic

Vertex 164
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-350.319, 74.218, 646.999, 24.982)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #246 key=178

Compton
Vertex 165
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-372.320, 183.466, 563.448, 36.103)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #247 key=179

pi+ (211)
Particle 27
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.848, 0.144, 3.438, 3.909)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 58 simE=0.381807
direct recHits: 9 missing=49 recoE=3.765416
subgraph simHits: 58 simE=0.381807
subgraph recHits: 9 missing=49 recoE=3.765416
direct tracker simHits: 19 dE=0.002077
subgraph tracker simHits: 19 dE=0.002077
checkpointId: 0
x4@checkpoint: (-110.284, 55.139, 230.072, 0.000)
p4@checkpoint: (-1.242, 1.315, 3.393, 3.847)
raw GEN: GenParticle #44 key=29
raw SIM: SimTrack #276 key=13

pi- (-211)
Particle 28
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.935, 0.491, 7.209, 8.228)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 106 simE=2.343462
direct recHits: 8 missing=98 recoE=1.128489
subgraph simHits: 106 simE=2.343462
subgraph recHits: 8 missing=98 recoE=1.128489
direct tracker simHits: 21 dE=0.002261
subgraph tracker simHits: 21 dE=0.002261
checkpointId: 0
x4@checkpoint: (-123.128, -6.505, 220.759, 0.000)
p4@checkpoint: (-3.849, -0.892, 7.179, 8.195)
raw GEN: GenParticle #45 key=30
raw SIM: SimTrack #277 key=14

nu_tau (16)
Particle 29
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (60.116, -8.695, -113.473, 128.708)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #46 key=31
raw SIM: n/a

pi- (-211)
Particle 30
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (3.769, -0.322, -6.347, 7.390)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 137 simE=4.101648
direct recHits: 0 missing=137 recoE=0.000000
subgraph simHits: 141 simE=4.106602
subgraph recHits: 0 missing=141 recoE=0.000000
direct tracker simHits: 19 dE=0.001894
subgraph tracker simHits: 19 dE=0.001894
checkpointId: 0
x4@checkpoint: (122.727, 11.877, -212.415, 0.000)
p4@checkpoint: (3.604, 1.041, -6.308, 7.340)
raw GEN: GenParticle #47 key=32
raw SIM: SimTrack #278 key=16

HadronInelastic

Vertex 119
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.393, 15.903, -231.215, 10.430)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #200 key=132

HadronInelastic

Vertex 120
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-107.626, 35.255, -136.998, 64.569)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #201 key=133

nu_tau (16)
Particle 31
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-10.247, -5.038, -4.400, 12.236)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #48 key=33
raw SIM: n/a

mu- (13)
Particle 32
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu- (13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-142.516, -71.716, -58.583, 169.959)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.354602
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.354602
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 15 dE=0.001331
subgraph tracker simHits: 15 dE=0.001331
checkpointId: 0
x4@checkpoint: (-109.899, -55.903, -49.772, 0.000)
p4@checkpoint: (-141.856, -72.954, -58.587, 169.935)
raw GEN: GenParticle #49 key=34
raw SIM: SimTrack #279 key=18

anti-nu_mu (-14)
Particle 33
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_mu (-14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-2.588, -1.259, -1.276, 3.148)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #50 key=35
raw SIM: n/a

anti-nu_tau (-16)
Particle 34
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (24.821, -21.896, -139.912, 143.774)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #51 key=36
raw SIM: n/a

pi+ (211)
Particle 35
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (10.891, -9.608, -66.658, 68.222)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 341 simE=0.054505
direct recHits: 257 missing=84 recoE=35.930195
subgraph simHits: 341 simE=0.054505
subgraph recHits: 257 missing=84 recoE=35.930195
direct tracker simHits: 19 dE=0.001694
subgraph tracker simHits: 19 dE=0.001694
checkpointId: 0
x4@checkpoint: (46.187, -42.752, -293.500, 0.000)
p4@checkpoint: (10.410, -10.090, -66.629, 68.188)
raw GEN: GenParticle #52 key=37
raw SIM: SimTrack #280 key=21

HadronInelastic

Vertex 112
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (56.552, -52.945, -360.470, 12.319)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #193 key=125

nu_tau (16)
Particle 36
stableLegsFromUpstream
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-19.676, 18.139, 118.930, 121.904)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #53 key=38
raw SIM: n/a

pi- (-211)
Particle 37
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-16.036, 13.365, 87.640, 90.092)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 1105 simE=0.419037
direct recHits: 872 missing=233 recoE=91.599129
subgraph simHits: 1115 simE=0.421371
subgraph recHits: 872 missing=243 recoE=91.599129
direct tracker simHits: 30 dE=0.003417
subgraph tracker simHits: 30 dE=0.003417
checkpointId: 0
x4@checkpoint: (-55.314, 44.373, 293.500, 0.000)
p4@checkpoint: (-16.499, 12.736, 87.605, 90.051)
raw GEN: GenParticle #54 key=39
raw SIM: SimTrack #281 key=23

HadronInelastic

Vertex 111
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-75.404, 69.716, 268.008, 27.107)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #192 key=124

anti-nu_tau (-16)
Particle 38
stableLegsFromUpstream
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (52.860, 26.532, 22.247, 63.190)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #55 key=40
raw SIM: n/a

e+ (-11)
Particle 39
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (101.861, 51.186, 41.648, 121.368)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 157 simE=44.378357
direct recHits: 0 missing=157 recoE=0.000000
subgraph simHits: 280 simE=120.101166
subgraph recHits: 0 missing=280 recoE=0.000000
direct tracker simHits: 16 dE=0.001446
subgraph tracker simHits: 16 dE=0.001446
checkpointId: 0
x4@checkpoint: (110.513, 54.678, 40.554, 0.000)
p4@checkpoint: (37.895, 18.396, 15.386, 44.846)
raw GEN: GenParticle #56 key=41
raw SIM: SimTrack #282 key=25

Bremsstrahlung
Vertex 96
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (44.912, 22.430, 13.852, 1.950)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #177 key=109

Bremsstrahlung
Vertex 97
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (63.320, 31.543, 21.355, 2.679)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #178 key=110

Bremsstrahlung
Vertex 98
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.286, 51.160, 37.618, 4.260)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #179 key=111

Bremsstrahlung
Vertex 99
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.498, 51.263, 37.704, 4.269)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #180 key=112

Bremsstrahlung
Vertex 100
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.565, 51.296, 37.731, 4.271)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #181 key=113

Bremsstrahlung
Vertex 101
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.802, 51.412, 37.827, 4.281)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #182 key=114

Bremsstrahlung
Vertex 102
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.293, 51.651, 38.027, 4.300)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #183 key=115

Bremsstrahlung
Vertex 103
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.555, 51.779, 38.133, 4.310)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #184 key=116

Bremsstrahlung
Vertex 104
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.150, 52.556, 38.781, 4.373)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #185 key=117

nu_e (12)
Particle 40
stableLegsFromUpstream
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.631, 0.295, 0.364, 0.785)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #57 key=42
raw SIM: n/a

gamma (22)
Particle 41
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (8.575, 2.313, 6.155, 10.806)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 97 simE=10.666600
subgraph recHits: 0 missing=97 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.046, 32.108, 80.952, 0.000)
p4@checkpoint: (8.575, 2.313, 6.155, 10.806)
raw GEN: GenParticle #58 key=43
raw SIM: SimTrack #283 key=27

PairConversion

Vertex 95
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (125.559, 33.865, 85.628, 5.445)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #176 key=108

gamma (22)
Particle 42
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (2.822, 0.698, 1.957, 3.505)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 57 simE=3.460066
subgraph recHits: 0 missing=57 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.687, 29.630, 78.506, 0.000)
p4@checkpoint: (2.822, 0.698, 1.957, 3.505)
raw GEN: GenParticle #59 key=44
raw SIM: SimTrack #284 key=28

PairConversion

Vertex 94
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (125.812, 31.146, 82.754, 5.379)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #175 key=107

gamma (22)
Particle 43
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.931, -0.637, -1.334, 2.432)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 50 simE=2.379171
subgraph recHits: 0 missing=50 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.106, -38.588, -85.378, 0.000)
p4@checkpoint: (-1.931, -0.637, -1.334, 2.432)
raw GEN: GenParticle #60 key=45
raw SIM: SimTrack #285 key=29

PairConversion

Vertex 93
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-125.062, -41.211, -90.873, 5.420)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #174 key=106

gamma (22)
Particle 44
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-7.758, -2.278, -5.359, 9.700)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 227 simE=9.052075
subgraph recHits: 0 missing=227 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 71 dE=0.013791
raw GEN: GenParticle #61 key=46
raw SIM: SimTrack #286 key=30

PairConversion

Vertex 39
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-5.739, -1.678, -8.459, 0.406)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #111 key=43

gamma (22)
Particle 45
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (39.405, -31.611, 38.089, 63.268)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 213 simE=62.274643
subgraph recHits: 0 missing=213 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.176, -77.156, 88.475, 0.000)
p4@checkpoint: (39.405, -31.611, 38.089, 63.268)
raw GEN: GenParticle #62 key=47
raw SIM: SimTrack #287 key=31

PairConversion

Vertex 36
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.905, -80.950, 93.046, 5.571)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #108 key=40

gamma (22)
Particle 46
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (7.107, -5.632, 6.813, 11.342)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 110 simE=11.182560
subgraph recHits: 0 missing=110 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.630, -76.586, 88.143, 0.000)
p4@checkpoint: (7.107, -5.632, 6.813, 11.342)
raw GEN: GenParticle #63 key=48
raw SIM: SimTrack #288 key=32

PairConversion

Vertex 35
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.729, -80.627, 93.031, 5.583)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #107 key=39

gamma (22)
Particle 47
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-16.752, 13.933, -16.754, 27.486)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 169 simE=27.175259
subgraph recHits: 0 missing=169 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #64 key=49
raw SIM: SimTrack #289 key=33

PairConversion

Vertex 31
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.649, 76.225, -96.149, 5.183)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #103 key=35

gamma (22)
Particle 48
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-20.473, 17.077, -20.318, 33.520)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 155 simE=33.202503
subgraph recHits: 0 missing=155 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.687, 78.977, -98.457, 0.000)
p4@checkpoint: (-20.473, 17.077, -20.318, 33.520)
raw GEN: GenParticle #65 key=50
raw SIM: SimTrack #290 key=34

PairConversion

Vertex 29
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.904, 83.328, -103.634, 5.623)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #100 key=32

gamma (22)
Particle 49
stableLegsFromUpstream
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-13.551, 1.871, 25.537, 28.970)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 149 simE=27.880678
subgraph recHits: 0 missing=149 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.141, 16.863, 225.679, 0.000)
p4@checkpoint: (-13.551, 1.871, 25.537, 28.970)
raw GEN: GenParticle #66 key=51
raw SIM: SimTrack #291 key=35

PairConversion

Vertex 27
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-132.001, 18.224, 244.258, 9.580)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #98 key=30

gamma (22)
Particle 50
stableLegsFromUpstream
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-5.290, 0.687, 10.113, 11.433)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 157 simE=11.009885
subgraph recHits: 0 missing=157 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #67 key=52
raw SIM: SimTrack #292 key=36

PairConversion

Vertex 15
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.857, 14.915, 215.070, 8.447)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #83 key=15

e- (11)
Particle 51
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-4.093, 0.532, 7.825, 8.847)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 75 simE=3.575964
direct recHits: 0 missing=75 recoE=0.000000
subgraph simHits: 128 simE=8.603331
subgraph recHits: 0 missing=128 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.288, 15.766, 229.259, 0.000)
p4@checkpoint: (-1.705, 0.156, 3.244, 3.668)
raw GEN: n/a
raw SIM: SimTrack #293 key=37

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.224, 14.963, 215.773, 8.474)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #92 key=24

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.725, 15.029, 216.731, 8.510)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #93 key=25

Bremsstrahlung
Vertex 24
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.776, 15.036, 216.830, 8.514)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #94 key=26

Bremsstrahlung
Vertex 25
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.711, 15.156, 218.620, 8.581)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #95 key=27

Bremsstrahlung
Vertex 26
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.730, 15.158, 218.657, 8.583)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #96 key=28

e+ (-11)
Particle 52
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.197, 0.155, 2.288, 2.587)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 53 simE=1.380557
direct recHits: 0 missing=53 recoE=0.000000
subgraph simHits: 68 simE=2.406554
subgraph recHits: 0 missing=68 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.221, 16.274, 229.282, 0.000)
p4@checkpoint: (-0.647, 0.159, 1.264, 1.429)
raw GEN: n/a
raw SIM: SimTrack #294 key=38

Bremsstrahlung
Vertex 16
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.866, 14.916, 215.088, 8.448)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #84 key=16

Bremsstrahlung
Vertex 17
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.427, 14.990, 216.161, 8.488)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #85 key=17

Bremsstrahlung
Vertex 18
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.517, 15.002, 216.335, 8.495)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #86 key=18

Bremsstrahlung
Vertex 19
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.084, 15.084, 217.420, 8.536)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #87 key=19

Bremsstrahlung
Vertex 20
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.710, 15.180, 218.621, 8.581)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #88 key=20

gamma (22)
Particle 53
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.000, 0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.005155
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.005155
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.273, 15.881, 229.276, 0.000)
p4@checkpoint: (-0.003, 0.000, 0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #295 key=39

gamma (22)
Particle 54
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.001, 0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008032
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008032
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.268, 15.918, 229.281, 0.000)
p4@checkpoint: (-0.004, 0.001, 0.007, 0.008)
raw GEN: n/a
raw SIM: SimTrack #296 key=41

gamma (22)
Particle 55
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.068, 0.009, 0.130, 0.147)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 5 simE=0.090139
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 6 simE=0.090390
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.320, 15.515, 230.179, 0.000)
p4@checkpoint: (-0.040, 0.001, 0.085, 0.094)
raw GEN: n/a
raw SIM: SimTrack #297 key=42

Bremsstrahlung
Vertex 21
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-118.335, 14.187, 235.563, 9.305)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #90 key=22

gamma (22)
Particle 56
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.313, 0.047, 0.599, 0.678)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.641196
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=0.641196
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.257, 16.006, 229.261, 0.000)
p4@checkpoint: (-0.313, 0.047, 0.599, 0.678)
raw GEN: n/a
raw SIM: SimTrack #298 key=44

gamma (22)
Particle 57
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.140, 0.022, 0.270, 0.305)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.281225
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.281225
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.251, 16.052, 229.271, 0.000)
p4@checkpoint: (-0.140, 0.022, 0.270, 0.305)
raw GEN: n/a
raw SIM: SimTrack #299 key=45

gamma (22)
Particle 58
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000251
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000251
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.304, 15.644, 237.483, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #301 key=1739

gamma (22)
Particle 59
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-2.229, 0.292, 4.266, 4.822)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 90 simE=4.690689
direct recHits: 0 missing=90 recoE=0.000000
subgraph simHits: 90 simE=4.690689
subgraph recHits: 0 missing=90 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.272, 15.885, 229.260, 0.000)
p4@checkpoint: (-2.229, 0.292, 4.266, 4.822)
raw GEN: n/a
raw SIM: SimTrack #303 key=1756

gamma (22)
Particle 60
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.118, 0.016, 0.227, 0.256)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.253447
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.253447
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.271, 15.894, 229.257, 0.000)
p4@checkpoint: (-0.118, 0.016, 0.227, 0.256)
raw GEN: n/a
raw SIM: SimTrack #304 key=1758

gamma (22)
Particle 61
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000721
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000721
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.271, 15.892, 229.275, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #305 key=1759

gamma (22)
Particle 62
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.001, 0.008, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.009110
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.009110
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.277, 15.854, 229.291, 0.000)
p4@checkpoint: (-0.004, 0.001, 0.008, 0.009)
raw GEN: n/a
raw SIM: SimTrack #306 key=1760

gamma (22)
Particle 63
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.035, 0.004, 0.067, 0.076)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.073400
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.073400
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.277, 15.848, 229.288, 0.000)
p4@checkpoint: (-0.035, 0.004, 0.067, 0.076)
raw GEN: n/a
raw SIM: SimTrack #307 key=1761

e- (11)
Particle 64
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-13.222, 1.824, 24.914, 28.264)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 148 simE=27.186403
direct recHits: 0 missing=148 recoE=0.000000
subgraph simHits: 149 simE=27.186914
subgraph recHits: 0 missing=149 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-132.024, 18.227, 244.302, 0.000)
p4@checkpoint: (-10.958, 1.513, 20.653, 23.428)
raw GEN: n/a
raw SIM: SimTrack #310 key=7615

Annihilation

Vertex 28
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (32.261, 111.568, 191.210, 17.423)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #99 key=31

e+ (-11)
Particle 65
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.330, 0.046, 0.622, 0.706)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=0.693761
direct recHits: 0 missing=28 recoE=0.000000
subgraph simHits: 28 simE=0.693761
subgraph recHits: 0 missing=28 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-132.012, 18.226, 244.280, 0.000)
p4@checkpoint: (-0.331, 0.047, 0.620, 0.704)
raw GEN: n/a
raw SIM: SimTrack #311 key=7616

gamma (22)
Particle 66
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (35.481, 118.085, 190.067, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #312 key=26214

e- (11)
Particle 67
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-18.621, 15.531, -18.479, 30.487)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 145 simE=30.212317
direct recHits: 0 missing=145 recoE=0.000000
subgraph simHits: 147 simE=30.212883
subgraph recHits: 0 missing=147 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.933, 83.353, -103.663, 0.000)
p4@checkpoint: (-18.522, 15.443, -18.382, 30.323)
raw GEN: n/a
raw SIM: SimTrack #314 key=26308

Annihilation

Vertex 30
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-100.495, 57.594, -99.990, 6.746)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #101 key=33

e+ (-11)
Particle 68
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.852, 1.545, -1.839, 3.033)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 48 simE=2.989619
direct recHits: 0 missing=48 recoE=0.000000
subgraph simHits: 48 simE=2.989619
subgraph recHits: 0 missing=48 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.980, 83.392, -103.710, 0.000)
p4@checkpoint: (-1.852, 1.547, -1.835, 3.032)
raw GEN: n/a
raw SIM: SimTrack #315 key=26309

gamma (22)
Particle 69
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000422
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000422
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-103.924, 66.352, -79.226, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #316 key=38516

gamma (22)
Particle 70
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000143
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000143
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.418, -14.718, -133.818, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #317 key=38517

e- (11)
Particle 71
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-10.263, 8.534, -10.262, 16.836)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 134 simE=16.614290
direct recHits: 0 missing=134 recoE=0.000000
subgraph simHits: 137 simE=16.614895
subgraph recHits: 0 missing=137 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.802, 78.838, -99.299, 0.000)
p4@checkpoint: (-10.291, 8.497, -10.264, 16.835)
raw GEN: n/a
raw SIM: SimTrack #319 key=47931

Annihilation

Vertex 32
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-9.214, -51.216, -264.204, 13.419)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #104 key=36

Bremsstrahlung
Vertex 33
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.853, 80.522, -121.153, 6.482)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #105 key=37

Bremsstrahlung
Vertex 34
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.956, 81.090, -121.105, 6.527)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #106 key=38

e+ (-11)
Particle 72
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-6.490, 5.398, -6.492, 10.649)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 96 simE=10.560354
direct recHits: 0 missing=96 recoE=0.000000
subgraph simHits: 96 simE=10.560354
subgraph recHits: 0 missing=96 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.791, 78.852, -99.298, 0.000)
p4@checkpoint: (-6.461, 5.433, -6.490, 10.648)
raw GEN: n/a
raw SIM: SimTrack #320 key=47932

gamma (22)
Particle 73
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000110
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000110
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.859, -114.454, -193.552, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #322 key=62770

gamma (22)
Particle 74
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000167
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000167
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.374, 85.982, -130.297, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #323 key=65570

gamma (22)
Particle 75
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000327
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000327
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.493, -26.160, -98.063, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #324 key=65575

e- (11)
Particle 76
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (3.002, -2.379, 2.878, 4.791)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 78 simE=4.725166
direct recHits: 0 missing=78 recoE=0.000000
subgraph simHits: 78 simE=4.725166
subgraph recHits: 0 missing=78 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.741, -80.636, 93.042, 0.000)
p4@checkpoint: (3.002, -2.380, 2.876, 4.791)
raw GEN: n/a
raw SIM: SimTrack #325 key=65674

e+ (-11)
Particle 77
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (4.106, -3.253, 3.933, 6.550)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 80 simE=6.457396
direct recHits: 0 missing=80 recoE=0.000000
subgraph simHits: 80 simE=6.457396
subgraph recHits: 0 missing=80 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.971, -80.819, 93.263, 0.000)
p4@checkpoint: (3.618, -2.864, 3.464, 5.769)
raw GEN: n/a
raw SIM: SimTrack #326 key=65675

e- (11)
Particle 78
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (19.969, -16.020, 19.302, 32.062)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 168 simE=31.534170
direct recHits: 0 missing=168 recoE=0.000000
subgraph simHits: 168 simE=31.534170
subgraph recHits: 0 missing=168 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.928, -80.969, 93.068, 0.000)
p4@checkpoint: (19.967, -16.014, 19.299, 32.056)
raw GEN: n/a
raw SIM: SimTrack #327 key=72993

e+ (-11)
Particle 79
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (19.436, -15.592, 18.787, 31.206)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 132 simE=30.739882
direct recHits: 0 missing=132 recoE=0.000000
subgraph simHits: 134 simE=30.740458
subgraph recHits: 0 missing=134 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.934, -80.973, 93.074, 0.000)
p4@checkpoint: (18.414, -14.777, 17.802, 29.570)
raw GEN: n/a
raw SIM: SimTrack #328 key=72994

Bremsstrahlung
Vertex 37
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.896, -112.783, 104.446, 10.387)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #109 key=41

Bremsstrahlung
Vertex 38
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (93.662, -67.765, 80.586, 6.670)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #110 key=42

gamma (22)
Particle 80
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000304
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000304
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-36.543, -117.760, 105.392, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #329 key=75202

gamma (22)
Particle 81
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000271
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000271
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.764, -63.379, 79.999, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #330 key=79157

e- (11)
Particle 82
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.630, -0.772, -1.815, 3.287)
nProdVtx: 1 nDecayVtx: 8
nParents: 1 nChildren: 8
nCheckpoints: 1
direct simHits: 47 simE=2.078479
direct recHits: 0 missing=47 recoE=0.000000
subgraph simHits: 72 simE=3.181925
subgraph recHits: 0 missing=72 recoE=0.000000
direct tracker simHits: 16 dE=0.002073
subgraph tracker simHits: 16 dE=0.002073
checkpointId: 0
x4@checkpoint: (-107.000, -61.269, -88.727, 0.000)
p4@checkpoint: (-1.221, -1.258, -1.175, 2.110)
raw GEN: n/a
raw SIM: SimTrack #331 key=113862

Bremsstrahlung
Vertex 85
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.387, -35.636, -61.496, 3.573)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #166 key=98

Bremsstrahlung
Vertex 86
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-100.945, -55.385, -83.056, 4.847)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #167 key=99

Bremsstrahlung
Vertex 87
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.281, -55.695, -83.367, 4.866)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #168 key=100

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.319, -55.730, -83.402, 4.868)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #169 key=101

Bremsstrahlung
Vertex 89
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.354, -55.763, -83.435, 4.870)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #170 key=102

Bremsstrahlung
Vertex 90
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.845, -56.217, -83.887, 4.897)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #171 key=103

Bremsstrahlung
Vertex 91
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-103.302, -57.588, -85.229, 4.977)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #172 key=104

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-103.682, -57.950, -85.581, 4.998)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #173 key=105

e+ (-11)
Particle 83
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-5.128, -1.506, -3.543, 6.413)
nProdVtx: 1 nDecayVtx: 17
nParents: 1 nChildren: 17
nCheckpoints: 1
direct simHits: 56 simE=2.151568
direct recHits: 0 missing=56 recoE=0.000000
subgraph simHits: 155 simE=5.870149
subgraph recHits: 0 missing=155 recoE=0.000000
direct tracker simHits: 16 dE=0.001539
subgraph tracker simHits: 63 dE=0.011718
checkpointId: 0
x4@checkpoint: (-122.069, -17.376, -86.625, 0.000)
p4@checkpoint: (-1.816, 0.109, -1.199, 2.179)
raw GEN: n/a
raw SIM: SimTrack #332 key=113863

Bremsstrahlung
Vertex 40
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.105, -5.648, -18.344, 1.003)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #112 key=44

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.906, -6.112, -19.578, 1.077)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #113 key=45

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-37.958, -9.857, -30.505, 1.737)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #114 key=46

Bremsstrahlung
Vertex 43
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-38.117, -9.891, -30.613, 1.744)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #115 key=47

Bremsstrahlung
Vertex 44
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-50.475, -12.297, -38.950, 2.247)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #116 key=48

Bremsstrahlung
Vertex 45
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-53.561, -12.834, -41.025, 2.373)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #117 key=49

Bremsstrahlung
Vertex 46
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-69.590, -15.173, -51.748, 3.021)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #118 key=50

Bremsstrahlung
Vertex 47
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-113.665, -17.652, -81.069, 4.789)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #119 key=51

Bremsstrahlung
Vertex 48
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.084, -17.648, -81.346, 4.806)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #120 key=52

Bremsstrahlung
Vertex 49
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.131, -17.647, -81.377, 4.808)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #121 key=53

Bremsstrahlung
Vertex 50
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.157, -17.647, -81.395, 4.809)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #122 key=54

Bremsstrahlung
Vertex 51
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.229, -17.646, -81.442, 4.812)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #123 key=55

Bremsstrahlung
Vertex 52
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.326, -17.645, -81.506, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #124 key=56

Bremsstrahlung
Vertex 53
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.339, -17.645, -81.515, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #125 key=57

Bremsstrahlung
Vertex 54
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.379, -17.645, -81.542, 4.818)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #126 key=58

Bremsstrahlung
Vertex 55
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.974, -17.637, -81.937, 4.842)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #127 key=59

Bremsstrahlung
Vertex 56
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-116.724, -17.605, -83.096, 4.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #128 key=60

gamma (22)
Particle 84
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.167, -0.043, -0.114, 0.207)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 21 simE=0.017227
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 44 dE=0.007853
subgraph tracker simHits: 49 dE=0.010179
raw GEN: n/a
raw SIM: SimTrack #333 key=113865

PairConversion

Vertex 67
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-52.320, -14.003, -40.424, 2.335)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #141 key=73

gamma (22)
Particle 85
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.582, -0.149, -0.399, 0.721)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 37 simE=0.630896
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #334 key=113866

PairConversion

Vertex 62
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.347, -29.973, -83.559, 4.938)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #134 key=66

gamma (22)
Particle 86
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001630
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001630
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.248, -27.265, -86.194, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #335 key=113868

gamma (22)
Particle 87
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.013, -0.003, -0.009, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.012710
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.012710
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.251, -27.252, -86.212, 0.000)
p4@checkpoint: (-0.013, -0.003, -0.009, 0.016)
raw GEN: n/a
raw SIM: SimTrack #336 key=113869

gamma (22)
Particle 88
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.496, -0.088, -0.334, 0.604)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 22 simE=0.600293
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 22 simE=0.600293
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.776, -24.822, -86.237, 0.000)
p4@checkpoint: (-0.496, -0.088, -0.334, 0.604)
raw GEN: n/a
raw SIM: SimTrack #337 key=113870

gamma (22)
Particle 89
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, -0.004, -0.014, 0.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.026168
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.026168
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.895, -24.233, -86.313, 0.000)
p4@checkpoint: (-0.022, -0.004, -0.014, 0.026)
raw GEN: n/a
raw SIM: SimTrack #338 key=113872

gamma (22)
Particle 90
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.115, -0.014, -0.077, 0.139)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.005361
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.005361
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.418, -21.458, -86.371, 0.000)
p4@checkpoint: (-0.115, -0.014, -0.077, 0.139)
raw GEN: n/a
raw SIM: SimTrack #339 key=113875

gamma (22)
Particle 91
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.120, 0.001, -0.079, 0.144)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.142840
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.142840
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.040, -17.581, -86.624, 0.000)
p4@checkpoint: (-0.120, 0.001, -0.079, 0.144)
raw GEN: n/a
raw SIM: SimTrack #340 key=113877

gamma (22)
Particle 92
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002360
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002360
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.042, -17.567, -86.623, 0.000)
p4@checkpoint: (-0.002, 0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #341 key=113878

gamma (22)
Particle 93
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.095, 0.001, -0.063, 0.115)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 12 simE=0.078873
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 18 simE=0.085563
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.042, -17.567, -86.622, 0.000)
p4@checkpoint: (-0.095, 0.001, -0.063, 0.115)
raw GEN: n/a
raw SIM: SimTrack #342 key=113879

Bremsstrahlung
Vertex 57
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (31.431, 111.600, -87.438, 12.335)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #129 key=61

Bremsstrahlung
Vertex 58
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (32.817, 111.130, -87.840, 12.388)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #130 key=62

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (32.964, 110.921, -87.825, 12.397)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #131 key=63

Bremsstrahlung
Vertex 60
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (31.427, 111.652, -87.424, 12.336)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #132 key=64

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (31.457, 111.780, -87.424, 12.340)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #133 key=65

gamma (22)
Particle 94
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, 0.000, -0.010, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.017783
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.017783
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.043, -17.563, -86.619, 0.000)
p4@checkpoint: (-0.015, 0.000, -0.010, 0.018)
raw GEN: n/a
raw SIM: SimTrack #343 key=113880

gamma (22)
Particle 95
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.422, 0.005, -0.280, 0.507)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 26 simE=0.504868
direct recHits: 0 missing=26 recoE=0.000000
subgraph simHits: 26 simE=0.504868
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.043, -17.558, -86.628, 0.000)
p4@checkpoint: (-0.422, 0.005, -0.280, 0.507)
raw GEN: n/a
raw SIM: SimTrack #344 key=113881

gamma (22)
Particle 96
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, 0.000, -0.007, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.013145
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.013145
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.042, -17.565, -86.629, 0.000)
p4@checkpoint: (-0.011, 0.000, -0.007, 0.013)
raw GEN: n/a
raw SIM: SimTrack #345 key=113882

gamma (22)
Particle 97
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, 0.000, -0.014, 0.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.025971
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.025971
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.043, -17.561, -86.631, 0.000)
p4@checkpoint: (-0.022, 0.000, -0.014, 0.026)
raw GEN: n/a
raw SIM: SimTrack #346 key=113883

gamma (22)
Particle 98
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.480, 0.005, -0.319, 0.576)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.562258
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=0.562258
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.042, -17.565, -86.626, 0.000)
p4@checkpoint: (-0.480, 0.005, -0.319, 0.576)
raw GEN: n/a
raw SIM: SimTrack #347 key=113884

gamma (22)
Particle 99
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.894, 0.012, -0.594, 1.073)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 32 simE=1.061299
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 32 simE=1.061299
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.046, -17.538, -86.635, 0.000)
p4@checkpoint: (-0.894, 0.012, -0.594, 1.073)
raw GEN: n/a
raw SIM: SimTrack #348 key=113885

gamma (22)
Particle 100
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.000, -0.005, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.008209
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.008209
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.055, -17.478, -86.626, 0.000)
p4@checkpoint: (-0.007, 0.000, -0.005, 0.008)
raw GEN: n/a
raw SIM: SimTrack #349 key=113887

gamma (22)
Particle 101
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000283
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000283
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.856, 113.208, -92.814, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #350 key=116790

gamma (22)
Particle 102
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.003, -0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003922
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003922
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.264, -15.952, -125.818, 0.000)
p4@checkpoint: (0.002, -0.003, -0.001, 0.004)
raw GEN: n/a
raw SIM: SimTrack #351 key=116794

gamma (22)
Particle 103
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.002, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001080
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001080
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.356, -40.794, -25.276, 0.000)
p4@checkpoint: (0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #352 key=116798

gamma (22)
Particle 104
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000567
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000567
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (35.067, 118.208, -86.947, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #353 key=116813

gamma (22)
Particle 105
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000837
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000837
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (31.882, 119.107, -87.814, 0.000)
p4@checkpoint: (0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #354 key=116814

e- (11)
Particle 106
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.524, -0.135, -0.359, 0.649)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 30 simE=0.436181
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 34 simE=0.630268
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.280, -31.228, -86.312, 0.000)
p4@checkpoint: (-0.350, -0.136, -0.251, 0.452)
raw GEN: n/a
raw SIM: SimTrack #355 key=117410

Bremsstrahlung
Vertex 66
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.488, -30.009, -83.656, 4.944)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #140 key=72

e+ (-11)
Particle 107
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.058, -0.014, -0.040, 0.072)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000627
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.883, -28.826, -86.652, 0.000)
p4@checkpoint: (-0.045, 0.039, -0.038, 0.071)
raw GEN: n/a
raw SIM: SimTrack #356 key=117411

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-113.605, -21.661, -97.129, 5.771)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #135 key=67

Annihilation

Vertex 64
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-113.073, -22.388, -98.595, 5.828)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #137 key=69

Annihilation

Vertex 65
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-104.832, -50.627, -148.588, 7.803)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #139 key=71

gamma (22)
Particle 108
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000280
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000280
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.239, -22.449, -89.053, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #357 key=117418

gamma (22)
Particle 109
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000274
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000274
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.428, -34.317, -100.263, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #359 key=117423

gamma (22)
Particle 110
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000074
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000074
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.505, -30.355, -159.079, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #361 key=117434

gamma (22)
Particle 111
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.158, -0.041, -0.108, 0.196)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.194087
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.194087
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.335, -31.016, -86.285, 0.000)
p4@checkpoint: (-0.158, -0.041, -0.108, 0.196)
raw GEN: n/a
raw SIM: SimTrack #362 key=117484

e- (11)
Particle 112
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.067, -0.015, -0.045, 0.082)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.006668
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 8 dE=0.001065
subgraph tracker simHits: 8 dE=0.001065
checkpointId: 0
x4@checkpoint: (-50.831, -20.646, -293.500, 0.000)
p4@checkpoint: (-0.002, 0.005, -0.008, 0.010)
raw GEN: n/a
raw SIM: SimTrack #363 key=117903

Bremsstrahlung
Vertex 79
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-51.346, -25.613, -52.230, 3.040)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #155 key=87

Bremsstrahlung
Vertex 80
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-53.860, -18.451, -69.584, 3.930)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #156 key=88

Bremsstrahlung
Vertex 81
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-50.308, -22.898, -97.607, 5.351)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #157 key=89

Bremsstrahlung
Vertex 82
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-50.222, -19.485, -155.626, 7.954)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #159 key=91

Annihilation

Vertex 83
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (116.190, -12.545, -219.256, 12.250)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #164 key=96

Bremsstrahlung
Vertex 84
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (116.119, -12.359, -219.299, 12.238)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #165 key=97

e+ (-11)
Particle 113
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.100, -0.027, -0.069, 0.125)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 10
nCheckpoints: 1
direct simHits: 1 simE=0.000104
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 14 simE=0.010559
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 11 dE=0.001261
subgraph tracker simHits: 11 dE=0.001261
checkpointId: 0
x4@checkpoint: (-61.452, -5.571, -293.500, 0.000)
p4@checkpoint: (0.022, 0.016, -0.014, 0.031)
raw GEN: n/a
raw SIM: SimTrack #364 key=117904

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-59.893, -8.585, -222.460, 24.768)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #144 key=76

Annihilation

Vertex 69
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-59.642, -7.303, -222.465, 24.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #145 key=77

Bremsstrahlung
Vertex 71
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.168, 72.542, -177.986, 10.705)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #147 key=79

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.180, 72.575, -177.996, 10.706)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #148 key=80

Bremsstrahlung
Vertex 73
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.202, 73.058, -178.310, 10.725)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #149 key=81

Annihilation

Vertex 74
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.216, 72.806, -178.824, 10.794)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #150 key=82

Bremsstrahlung
Vertex 75
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.204, 72.529, -178.033, 10.705)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #151 key=83

Bremsstrahlung
Vertex 76
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.262, 69.919, -181.448, 10.970)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #152 key=84

Bremsstrahlung
Vertex 77
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.284, 70.650, -183.189, 11.049)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #153 key=85

Bremsstrahlung
Vertex 78
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.321, 70.659, -183.212, 11.051)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #154 key=86

gamma (22)
Particle 114
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000400
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000400
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-86.859, 87.512, -209.722, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #367 key=117932

gamma (22)
Particle 115
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.003, 0.001, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.001017
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 2 simE=0.001186
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.914, 49.530, -199.941, 0.000)
p4@checkpoint: (0.001, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #368 key=117934

Bremsstrahlung
Vertex 70
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.600, 49.374, -196.703, 30.798)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #146 key=78

gamma (22)
Particle 116
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000169
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000169
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.578, 50.290, -197.226, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #370 key=117961

gamma (22)
Particle 117
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000185
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000185
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.867, 83.343, -181.243, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #371 key=117972

gamma (22)
Particle 118
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000617
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000617
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.749, 81.244, -180.457, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #372 key=117973

gamma (22)
Particle 119
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002285
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002285
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.833, 86.535, -185.694, 0.000)
p4@checkpoint: (0.000, 0.002, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #373 key=117974

gamma (22)
Particle 120
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001130
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001130
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.349, -122.945, -202.735, 0.000)
p4@checkpoint: (0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #374 key=117977

gamma (22)
Particle 121
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.003, -0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003593
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003593
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.699, 78.962, -182.037, 0.000)
p4@checkpoint: (-0.002, 0.003, -0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #376 key=117983

gamma (22)
Particle 122
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000494
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000494
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.158, 27.658, -278.490, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #377 key=117988

gamma (22)
Particle 123
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000454
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000454
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.736, 80.102, -181.471, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #378 key=117996

gamma (22)
Particle 124
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000113
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000113
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.933, 120.955, -149.511, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #379 key=117998

gamma (22)
Particle 125
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.000, -0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003630
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003630
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.640, -12.742, -169.037, 0.000)
p4@checkpoint: (0.003, 0.000, -0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #380 key=118006

gamma (22)
Particle 126
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000148
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000148
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.299, -0.450, -73.085, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #381 key=118009

gamma (22)
Particle 127
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000070
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000070
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.298, 0.730, -1.755, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #382 key=118012

gamma (22)
Particle 128
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001499
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001499
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.986, 23.777, -250.723, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #384 key=118020

gamma (22)
Particle 129
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.329, -15.445, -218.105, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #389 key=118047

gamma (22)
Particle 130
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000810
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000810
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.933, -9.513, -223.542, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #391 key=118051

gamma (22)
Particle 131
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000198
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000198
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-108.682, -58.234, -88.447, 0.000)
p4@checkpoint: (-0.002, -0.001, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #392 key=118059

gamma (22)
Particle 132
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001114
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001114
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.119, -61.060, -88.784, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #393 key=118062

gamma (22)
Particle 133
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.214, -0.199, -0.199, 0.354)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.352524
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.352524
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.103, -61.089, -88.776, 0.000)
p4@checkpoint: (-0.214, -0.199, -0.199, 0.354)
raw GEN: n/a
raw SIM: SimTrack #394 key=118063

gamma (22)
Particle 134
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000676
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000676
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.103, -61.088, -88.753, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #395 key=118064

gamma (22)
Particle 135
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.051, -0.047, -0.048, 0.085)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.065571
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.065571
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.111, -61.075, -88.759, 0.000)
p4@checkpoint: (-0.051, -0.047, -0.048, 0.085)
raw GEN: n/a
raw SIM: SimTrack #396 key=118065

gamma (22)
Particle 136
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.160, -0.149, -0.147, 0.263)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.259245
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.259245
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.095, -61.102, -88.706, 0.000)
p4@checkpoint: (-0.160, -0.149, -0.147, 0.263)
raw GEN: n/a
raw SIM: SimTrack #397 key=118067

gamma (22)
Particle 137
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.073, -0.069, -0.067, 0.121)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.115979
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.115979
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.063, -61.159, -88.713, 0.000)
p4@checkpoint: (-0.073, -0.069, -0.067, 0.121)
raw GEN: n/a
raw SIM: SimTrack #398 key=118069

gamma (22)
Particle 138
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.188, -0.182, -0.175, 0.315)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 15 simE=0.308138
direct recHits: 0 missing=15 recoE=0.000000
subgraph simHits: 15 simE=0.308138
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.035, -61.207, -88.715, 0.000)
p4@checkpoint: (-0.188, -0.182, -0.175, 0.315)
raw GEN: n/a
raw SIM: SimTrack #399 key=118070

e- (11)
Particle 139
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.596, -0.525, -1.102, 2.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 47 simE=1.961168
direct recHits: 0 missing=47 recoE=0.000000
subgraph simHits: 47 simE=1.961168
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-125.067, -41.213, -90.876, 0.000)
p4@checkpoint: (-1.293, -0.425, -0.894, 1.629)
raw GEN: n/a
raw SIM: SimTrack #400 key=120213

e+ (-11)
Particle 140
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.336, -0.111, -0.232, 0.423)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.418004
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.418004
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-125.097, -41.223, -90.897, 0.000)
p4@checkpoint: (-0.242, -0.079, -0.164, 0.303)
raw GEN: n/a
raw SIM: SimTrack #401 key=120214

e- (11)
Particle 141
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.776, 0.688, 1.925, 3.448)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 57 simE=3.403552
direct recHits: 0 missing=57 recoE=0.000000
subgraph simHits: 57 simE=3.403552
subgraph recHits: 0 missing=57 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (125.924, 31.173, 82.831, 0.000)
p4@checkpoint: (2.770, 0.684, 1.920, 3.439)
raw GEN: n/a
raw SIM: SimTrack #402 key=121735

e+ (-11)
Particle 142
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.046, 0.011, 0.032, 0.057)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.056515
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.056515
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (125.883, 31.163, 82.797, 0.000)
p4@checkpoint: (0.049, 0.015, 0.021, 0.055)
raw GEN: n/a
raw SIM: SimTrack #403 key=121736

e- (11)
Particle 143
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.267, 0.612, 1.627, 2.857)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 57 simE=2.827070
direct recHits: 0 missing=57 recoE=0.000000
subgraph simHits: 57 simE=2.827070
subgraph recHits: 0 missing=57 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (125.582, 33.871, 85.644, 0.000)
p4@checkpoint: (2.267, 0.614, 1.625, 2.856)
raw GEN: n/a
raw SIM: SimTrack #404 key=123944

e+ (-11)
Particle 144
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (6.308, 1.701, 4.528, 7.949)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 88 simE=7.839528
direct recHits: 0 missing=88 recoE=0.000000
subgraph simHits: 88 simE=7.839528
subgraph recHits: 0 missing=88 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (125.583, 33.871, 85.644, 0.000)
p4@checkpoint: (6.290, 1.701, 4.516, 7.928)
raw GEN: n/a
raw SIM: SimTrack #405 key=123945

gamma (22)
Particle 145
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.994, 1.486, 1.221, 3.558)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 76 simE=3.503642
subgraph recHits: 0 missing=76 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #406 key=130933

PairConversion

Vertex 106
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.400, 51.461, 37.711, 4.269)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #187 key=119

gamma (22)
Particle 146
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.039, 0.019, 0.016, 0.047)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.046587
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.046587
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.448, 54.810, 40.553, 0.000)
p4@checkpoint: (0.039, 0.019, 0.016, 0.047)
raw GEN: n/a
raw SIM: SimTrack #407 key=130938

gamma (22)
Particle 147
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (3.386, 1.652, 1.376, 4.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 67 simE=3.987012
direct recHits: 0 missing=67 recoE=0.000000
subgraph simHits: 67 simE=3.987012
subgraph recHits: 0 missing=67 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.510, 54.685, 40.554, 0.000)
p4@checkpoint: (3.386, 1.652, 1.376, 4.011)
raw GEN: n/a
raw SIM: SimTrack #408 key=130941

gamma (22)
Particle 148
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.158, 1.053, 0.877, 2.557)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 50 simE=2.534913
direct recHits: 0 missing=50 recoE=0.000000
subgraph simHits: 50 simE=2.534913
subgraph recHits: 0 missing=50 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.510, 54.685, 40.554, 0.000)
p4@checkpoint: (2.158, 1.053, 0.877, 2.557)
raw GEN: n/a
raw SIM: SimTrack #409 key=130943

gamma (22)
Particle 149
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (5.174, 2.524, 2.103, 6.129)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 61 simE=6.101360
direct recHits: 0 missing=61 recoE=0.000000
subgraph simHits: 61 simE=6.101360
subgraph recHits: 0 missing=61 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.510, 54.684, 40.554, 0.000)
p4@checkpoint: (5.174, 2.524, 2.103, 6.129)
raw GEN: n/a
raw SIM: SimTrack #410 key=130944

gamma (22)
Particle 150
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.030, 0.014, 0.012, 0.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.034803
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.034803
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.510, 54.684, 40.554, 0.000)
p4@checkpoint: (0.030, 0.014, 0.012, 0.035)
raw GEN: n/a
raw SIM: SimTrack #411 key=130946

gamma (22)
Particle 151
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (50.751, 24.752, 20.630, 60.116)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 187 simE=59.499176
direct recHits: 0 missing=187 recoE=0.000000
subgraph simHits: 201 simE=59.500282
subgraph recHits: 0 missing=201 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.511, 54.684, 40.554, 0.000)
p4@checkpoint: (50.751, 24.752, 20.630, 60.116)
raw GEN: n/a
raw SIM: SimTrack #412 key=130948

HadronInelastic

Vertex 105
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.787, 57.016, 31.084, 10.267)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #186 key=118

gamma (22)
Particle 152
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.002, 0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003811
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003811
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.511, 54.683, 40.554, 0.000)
p4@checkpoint: (0.003, 0.002, 0.001, 0.004)
raw GEN: n/a
raw SIM: SimTrack #413 key=130949

gamma (22)
Particle 153
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.009, 0.004, 0.004, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.010435
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.010435
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.512, 54.681, 40.554, 0.000)
p4@checkpoint: (0.009, 0.004, 0.004, 0.010)
raw GEN: n/a
raw SIM: SimTrack #414 key=130950

gamma (22)
Particle 154
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.001, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000207
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000207
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.270, 82.902, -104.153, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #415 key=183743

n (2112)
Particle 155
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.063, -0.010, 0.027, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.000080
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.000080
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.507, 13.961, 35.785, 0.000)
p4@checkpoint: (0.021, -0.022, 0.005, 0.940)
raw GEN: n/a
raw SIM: SimTrack #416 key=183744

gamma (22)
Particle 156
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000821
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000821
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (102.925, 67.893, 25.647, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #417 key=183745

e- (11)
Particle 157
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.769, 1.374, 1.129, 3.291)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 26 simE=0.385902
direct recHits: 0 missing=26 recoE=0.000000
subgraph simHits: 61 simE=3.255857
subgraph recHits: 0 missing=61 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.130, 55.447, 40.603, 0.000)
p4@checkpoint: (0.298, 0.224, 0.138, 0.398)
raw GEN: n/a
raw SIM: SimTrack #418 key=206840

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.425, 51.474, 37.721, 4.270)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #189 key=121

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.704, 51.613, 37.835, 4.281)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #190 key=122

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (105.837, 52.719, 38.720, 4.366)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #191 key=123

e+ (-11)
Particle 158
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.225, 0.112, 0.092, 0.267)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 23 simE=0.244408
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 24 simE=0.247784
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (111.014, 53.655, 40.474, 0.000)
p4@checkpoint: (0.236, 0.023, 0.085, 0.252)
raw GEN: n/a
raw SIM: SimTrack #419 key=206841

Bremsstrahlung
Vertex 107
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.705, 51.615, 37.829, 4.281)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #188 key=120

gamma (22)
Particle 159
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.001, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003376
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003376
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.444, 54.818, 40.299, 0.000)
p4@checkpoint: (0.003, 0.001, 0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #420 key=206846

gamma (22)
Particle 160
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.009, 0.005, 0.004, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.010061
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.010061
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.385, 54.936, 40.564, 0.000)
p4@checkpoint: (0.009, 0.005, 0.004, 0.011)
raw GEN: n/a
raw SIM: SimTrack #421 key=207019

gamma (22)
Particle 161
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.543, 0.776, 0.639, 1.842)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 45 simE=1.838908
direct recHits: 0 missing=45 recoE=0.000000
subgraph simHits: 45 simE=1.838908
subgraph recHits: 0 missing=45 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.371, 54.964, 40.598, 0.000)
p4@checkpoint: (1.543, 0.776, 0.639, 1.842)
raw GEN: n/a
raw SIM: SimTrack #422 key=207021

gamma (22)
Particle 162
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.856, 0.455, 0.356, 1.032)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 36 simE=1.020985
direct recHits: 0 missing=36 recoE=0.000000
subgraph simHits: 36 simE=1.020985
subgraph recHits: 0 missing=36 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.306, 55.094, 40.580, 0.000)
p4@checkpoint: (0.856, 0.455, 0.356, 1.032)
raw GEN: n/a
raw SIM: SimTrack #423 key=207022

n (2112)
Particle 163
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.034, -0.055, -0.063, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.000707
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.000707
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.070, -7.526, 179.573, 0.000)
p4@checkpoint: (-0.034, -0.055, -0.063, 0.944)
raw GEN: n/a
raw SIM: SimTrack #424 key=293759

gamma (22)
Particle 164
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001627
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001627
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.864, 84.424, 258.833, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #426 key=293761

pi+ (211)
Particle 165
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.583, -3.195, -23.064, 23.559)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 7
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #429 key=316974

HadronInelastic

Vertex 113
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.511, -80.573, -555.650, 18.965)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #194 key=126

PairConversion

Vertex 115
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.450, -80.984, -571.542, 19.500)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #196 key=128

Decay
Vertex 117
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.550, -80.723, -556.105, 18.982)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #198 key=130

Decay
Vertex 118
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (92.258, -90.190, -606.652, 20.716)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #199 key=131

pi+ (211)
Particle 166
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.339, -0.396, -1.606, 1.694)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #430 key=365457

pi- (-211)
Particle 167
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.344, -0.336, -1.116, 1.223)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #431 key=365462

pi+ (211)
Particle 168
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.998, -0.475, -5.236, 5.626)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #432 key=365466

HadronInelastic

Vertex 114
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.500, -88.893, -631.316, 21.683)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #195 key=127

pi+ (211)
Particle 169
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.117, -0.107, 0.116, 0.241)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #433 key=365940

e- (11)
Particle 170
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.022, -0.006, -0.164, 0.165)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #434 key=371796

Compton
Vertex 116
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.567, -64.632, -617.460, 21.129)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #197 key=129

e- (11)
Particle 171
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #435 key=371868

pi- (-211)
Particle 172
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.017, -0.461, -0.781, 0.917)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #436 key=372285

pi- (-211)
Particle 173
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.451, -0.445, -3.469, 3.529)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #437 key=374903

pi+ (211)
Particle 174
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.453, -0.845, -3.368, 3.505)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #438 key=374904

n (2112)
Particle 175
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.275, -0.039, 0.001, 0.980)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000577
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000577
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.062, -17.430, -230.068, 0.000)
p4@checkpoint: (-0.275, -0.039, 0.001, 0.980)
raw GEN: n/a
raw SIM: SimTrack #439 key=381635

gamma (22)
Particle 176
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.003, 0.003, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004377
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004377
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (6.242, 123.142, -66.430, 0.000)
p4@checkpoint: (0.004, 0.003, 0.003, 0.006)
raw GEN: n/a
raw SIM: SimTrack #441 key=382599

pi- (-211)
Particle 177
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.955, 0.395, 1.892, 2.161)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #444 key=387624

pi- (-211)
Particle 178
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.776, 0.447, 0.849, 1.242)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #445 key=387681

HadronInelastic

Vertex 125
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-418.778, 123.544, 727.760, 28.935)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #207 key=139

Compton
Vertex 126
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-417.848, 125.282, 714.826, 29.549)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #208 key=140

pi+ (211)
Particle 179
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.304, 0.136, 0.471, 0.594)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #446 key=387683

e- (11)
Particle 180
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.002, 0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #447 key=388459

e+ (-11)
Particle 181
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, -0.004, 0.008, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #448 key=388460

e- (11)
Particle 182
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.003, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #449 key=388524

p (2212)
Particle 183
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.272, 0.171, -0.502, 1.111)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #450 key=389265

e- (11)
Particle 184
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #451 key=389292

e- (11)
Particle 185
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #452 key=394267

e+ (-11)
Particle 186
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #453 key=394268

e- (11)
Particle 187
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.134, 0.050, 0.263, 0.299)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #454 key=394356

e+ (-11)
Particle 188
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.400, 0.497, 2.711, 3.091)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 13
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #455 key=394357

Compton
Vertex 129
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-419.279, 49.575, 678.672, 28.313)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #211 key=143

PairConversion

Vertex 130
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-375.951, 56.611, 687.347, 26.911)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #212 key=144

Compton
Vertex 131
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-348.445, 82.558, 678.581, 25.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #213 key=145

Compton
Vertex 132
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-344.785, 82.708, 667.232, 25.519)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #214 key=146

PairConversion

Vertex 133
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-336.219, 79.617, 648.739, 24.831)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #215 key=147

Compton
Vertex 134
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-381.332, 113.185, 718.687, 27.825)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #216 key=148

Compton
Vertex 135
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-351.302, 84.875, 678.604, 25.961)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #217 key=149

Compton
Vertex 136
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-363.610, 92.730, 695.058, 26.695)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #218 key=150

PairConversion

Vertex 138
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-336.593, 79.731, 649.446, 24.858)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #220 key=152

PairConversion

Vertex 139
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-336.043, 79.538, 648.364, 24.817)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #221 key=153

e- (11)
Particle 189
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, -0.004, -0.002, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #456 key=394493

e- (11)
Particle 190
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.002, 0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #457 key=394532

e+ (-11)
Particle 191
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.016, -0.011, 0.008, 0.021)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #458 key=394533

e- (11)
Particle 192
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, 0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #459 key=394634

e- (11)
Particle 193
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.001, 0.014, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #460 key=394655

e- (11)
Particle 194
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, 0.006, 0.031, 0.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #461 key=394697

e- (11)
Particle 195
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.002, 0.007, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #462 key=394720

e- (11)
Particle 196
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.026, 0.007, 0.055, 0.061)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #463 key=395040

Ionisation

Vertex 137
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-385.613, 119.649, 714.822, 28.125)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #219 key=151

e- (11)
Particle 197
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.004, 0.008, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #464 key=395041

e- (11)
Particle 198
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, 0.001, 0.002, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #465 key=395124

e- (11)
Particle 199
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.020, 0.006, 0.035, 0.041)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #466 key=395221

e+ (-11)
Particle 200
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.040, 0.014, 0.079, 0.089)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #467 key=395222

e- (11)
Particle 201
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.386, 0.136, 0.753, 0.857)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #468 key=395344

PairConversion

Vertex 142
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-336.735, 79.785, 649.748, 24.870)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #224 key=156

PairConversion

Vertex 144
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-336.747, 79.777, 649.761, 24.870)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #226 key=158

e+ (-11)
Particle 202
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.030, 0.011, 0.063, 0.070)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #469 key=395345

Ionisation

Vertex 140
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-382.696, 79.165, 678.693, 28.124)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #222 key=154

Compton
Vertex 141
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-316.111, 79.143, 678.706, 26.044)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #223 key=155

e- (11)
Particle 203
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.003, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #470 key=395392

e- (11)
Particle 204
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #471 key=395427

e- (11)
Particle 205
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.062, 0.025, 0.124, 0.141)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #472 key=395791

PairConversion

Vertex 143
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-364.568, 109.555, 692.603, 26.880)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #225 key=157

e+ (-11)
Particle 206
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.058, 0.022, 0.118, 0.133)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #473 key=395792

e+ (-11)
Particle 207
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, 0.024, 0.019, 0.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #474 key=396059

e- (11)
Particle 208
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, 0.003, 0.018, 0.021)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #475 key=396310

e+ (-11)
Particle 209
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.140, 0.047, 0.280, 0.316)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #476 key=396311

e- (11)
Particle 210
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.199, -0.030, 0.288, 0.351)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #477 key=397474

pi- (-211)
Particle 211
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.290, 0.112, 0.830, 0.897)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #478 key=397620

HadronInelastic

Vertex 163
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-361.792, 80.403, 683.783, 26.299)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #245 key=177

e- (11)
Particle 212
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #479 key=397634

e- (11)
Particle 213
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.005, 0.002, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #480 key=397764

e- (11)
Particle 214
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.007, 0.011, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #481 key=397779

e+ (-11)
Particle 215
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.009, 0.008, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #482 key=397780

pi+ (211)
Particle 216
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.046, -0.318, 0.389, 0.524)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #483 key=397990

e- (11)
Particle 217
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.587, 0.084, 4.287, 5.008)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #484 key=399150

Ionisation

Vertex 152
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-377.877, 76.264, 694.399, 26.812)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #234 key=166

PairConversion

Vertex 153
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-351.321, 74.329, 648.904, 25.053)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #235 key=167

Compton
Vertex 155
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-374.184, 65.101, 681.097, 26.406)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #237 key=169

PairConversion

Vertex 156
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-370.257, 74.763, 679.767, 26.261)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #238 key=170

e- (11)
Particle 218
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.009, 0.005, 0.015, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #485 key=399332

e+ (-11)
Particle 219
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.188, 0.005, 0.311, 0.363)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #486 key=399774

Ionisation

Vertex 154
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-375.651, 71.495, 682.311, 26.437)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #236 key=168

e- (11)
Particle 220
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #487 key=399786

e- (11)
Particle 221
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #488 key=399928

e- (11)
Particle 222
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, 0.000, 0.021, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #489 key=399973

e+ (-11)
Particle 223
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.279, 0.006, 0.454, 0.533)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #490 key=399974

e+ (-11)
Particle 224
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.208, -0.004, 0.294, 0.360)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #491 key=400865

Ionisation

Vertex 158
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-378.377, 67.123, 678.586, 26.420)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #240 key=172

Compton
Vertex 159
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-379.411, 76.161, 686.313, 26.614)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #241 key=173

e- (11)
Particle 225
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.002, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #492 key=400875

e- (11)
Particle 226
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.002, 0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #493 key=401152

e- (11)
Particle 227
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.002, 0.004, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #494 key=401269

e+ (-11)
Particle 228
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.070, -0.008, 0.144, 0.161)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #495 key=401466

Ionisation

Vertex 162
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-418.816, 48.215, 695.037, 28.041)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #244 key=176

e- (11)
Particle 229
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.020, -0.003, 0.002, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #496 key=401533

pi+ (211)
Particle 230
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.081, -0.060, -0.120, 0.210)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #497 key=401862

p (2212)
Particle 231
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.272, 0.024, 0.189, 0.995)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #498 key=401863

pi- (-211)
Particle 232
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, -0.007, -0.093, 0.168)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #499 key=401864

pi+ (211)
Particle 233
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.293, -0.264, 0.836, 0.935)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #500 key=405773

e- (11)
Particle 234
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.001, -0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #501 key=407751

n (2112)
Particle 235
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.074, -0.007, -0.156, 0.955)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.006515
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.006515
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-49.866, -112.767, 21.112, 0.000)
p4@checkpoint: (-0.074, -0.007, -0.156, 0.955)
raw GEN: n/a
raw SIM: SimTrack #502 key=418979

gamma (22)
Particle 236
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, -0.000, -0.002, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006470
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006470
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-51.101, -112.212, 35.168, 0.000)
p4@checkpoint: (-0.006, -0.000, -0.002, 0.006)
raw GEN: n/a
raw SIM: SimTrack #503 key=418980

n (2112)
Particle 237
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.054, 0.106, 0.102, 0.953)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.011506
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.011506
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (114.298, -46.247, 183.663, 0.000)
p4@checkpoint: (0.035, 0.028, 0.148, 0.952)
raw GEN: n/a
raw SIM: SimTrack #504 key=420746

gamma (22)
Particle 238
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.001887
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.001887
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.025, -67.740, 136.660, 0.000)
p4@checkpoint: (0.002, -0.001, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #505 key=420747

gamma (22)
Particle 239
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, -0.002, -0.000, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.001735
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.001735
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (58.092, -108.758, 107.963, 0.000)
p4@checkpoint: (-0.002, -0.002, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #506 key=424724

n (2112)
Particle 240
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.030, -0.039, 0.042, 0.942)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.001119
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000390
checkpointId: 0
x4@checkpoint: (66.515, -103.820, 120.322, 0.000)
p4@checkpoint: (-0.030, -0.039, 0.042, 0.942)
raw GEN: n/a
raw SIM: SimTrack #507 key=424725

NuclearCapture
Vertex 169
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.127, -103.489, 141.751, 115.767)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #251 key=183

gamma (22)
Particle 241
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000501
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000501
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.535, -89.760, 115.531, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #508 key=424726

gamma (22)
Particle 242
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000286
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000286
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000390
subgraph tracker simHits: 1 dE=0.000390
checkpointId: 0
x4@checkpoint: (84.925, -55.504, 293.500, 0.000)
p4@checkpoint: (0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #509 key=424731

gamma (22)
Particle 243
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000834
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000834
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.902, -39.201, 168.642, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #511 key=424733

gamma (22)
Particle 244
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000290
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000290
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (13.067, 122.606, 137.471, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #512 key=424759

gamma (22)
Particle 245
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000981
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000981
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-0.935, 123.296, 96.588, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #513 key=424760

gamma (22)
Particle 246
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001583
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001583
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.660, -66.765, 80.295, 0.000)
p4@checkpoint: (0.001, 0.001, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #514 key=425782

gamma (22)
Particle 247
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000117
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000117
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.813, -78.825, 86.553, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #515 key=425783

gamma (22)
Particle 248
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.002847
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.002847
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (87.272, -87.101, 84.652, 0.000)
p4@checkpoint: (0.002, -0.002, 0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #516 key=425785

n (2112)
Particle 249
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.047, 0.015, 0.015, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.000087
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.000087
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.537, -69.951, 94.677, 0.000)
p4@checkpoint: (0.038, 0.005, 0.028, 0.941)
raw GEN: n/a
raw SIM: SimTrack #517 key=425786

gamma (22)
Particle 250
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000173
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000173
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (83.339, -90.871, 81.931, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #518 key=425787

gamma (22)
Particle 251
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000389
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000389
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.596, 74.039, 52.136, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #519 key=426598

gamma (22)
Particle 252
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000323
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000323
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (111.550, -52.531, 94.443, 0.000)
p4@checkpoint: (-0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #520 key=426599

gamma (22)
Particle 253
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000937
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000937
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.283, -2.050, 111.858, 0.000)
p4@checkpoint: (0.001, 0.001, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #522 key=427611

gamma (22)
Particle 254
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000313
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000313
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.885, -59.697, 135.630, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #523 key=428021

n (2112)
Particle 255
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, 0.042, -0.066, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000602
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000602
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (57.322, 109.165, 30.667, 0.000)
p4@checkpoint: (0.035, 0.024, -0.040, 0.941)
raw GEN: n/a
raw SIM: SimTrack #524 key=432578

gamma (22)
Particle 256
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.002, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000370
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000370
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.897, 64.803, 231.224, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #525 key=432579

mu- (13)
Particle 257
caloBoundary
pid: mu- (13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.373, -1.293, -2.359, 4.316)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.301080
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.301080
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 7 dE=0.000784
subgraph tracker simHits: 7 dE=0.000784
checkpointId: 0
x4@checkpoint: (-112.315, -50.874, -85.809, 0.000)
p4@checkpoint: (-2.929, -2.091, -2.346, 4.297)
raw GEN: n/a
raw SIM: SimTrack #527 key=436181

n (2112)
Particle 258
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.097, -0.041, 0.036, 0.946)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.008516
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.008516
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.963, -77.421, -170.375, 0.000)
p4@checkpoint: (0.103, -0.012, -0.035, 0.946)
raw GEN: n/a
raw SIM: SimTrack #528 key=438035

n (2112)
Particle 259
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.058, 0.141, 0.047, 0.953)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.006380
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.006380
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.673, 19.967, -98.930, 0.000)
p4@checkpoint: (-0.058, 0.141, 0.047, 0.953)
raw GEN: n/a
raw SIM: SimTrack #530 key=438177

gamma (22)
Particle 260
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002104
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002104
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.731, 11.831, -101.210, 0.000)
p4@checkpoint: (-0.002, -0.000, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #531 key=438178

gamma (22)
Particle 261
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000181
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000181
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.030, 17.652, -103.889, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #532 key=438179

n (2112)
Particle 262
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.026, -0.094, 0.089, 0.949)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.014690
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.014690
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (33.941, -118.536, -43.082, 0.000)
p4@checkpoint: (0.026, -0.094, 0.089, 0.949)
raw GEN: n/a
raw SIM: SimTrack #533 key=438986

gamma (22)
Particle 263
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.001, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000196
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000196
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-111.759, 52.086, -85.137, 0.000)
p4@checkpoint: (-0.000, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #534 key=441859

gamma (22)
Particle 264
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001154
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001154
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.184, 16.549, -88.142, 0.000)
p4@checkpoint: (-0.002, -0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #535 key=441860

gamma (22)
Particle 265
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000087
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000087
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (11.106, 122.799, -73.198, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #536 key=441861

n (2112)
Particle 266
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.033, -0.026, 0.043, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.000673
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.000673
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.825, -79.998, 101.739, 0.000)
p4@checkpoint: (0.033, -0.026, 0.043, 0.942)
raw GEN: n/a
raw SIM: SimTrack #539 key=441869

gamma (22)
Particle 267
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000936
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000936
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-113.815, -47.424, -74.789, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #540 key=442294

gamma (22)
Particle 268
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.000, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002050
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002050
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.268, -2.830, -125.684, 0.000)
p4@checkpoint: (-0.003, 0.000, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #541 key=442298

connector

Particle 269
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 10
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4110 simE=386.056000
subgraph recHits: 1151 missing=2959 recoE=132.936539
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 312 dE=0.044670
raw GEN: n/a
raw SIM: n/a

ISR/upstream
Vertex 186
ISR/upstream (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.002, -0.000, -4.491, 0.167)
nIn: 1 nOut: 10
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 185
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (0.002, -0.000, -4.491, 0.167)
nIn: 0 nOut: 1
raw GEN: n/a
raw SIM: n/a